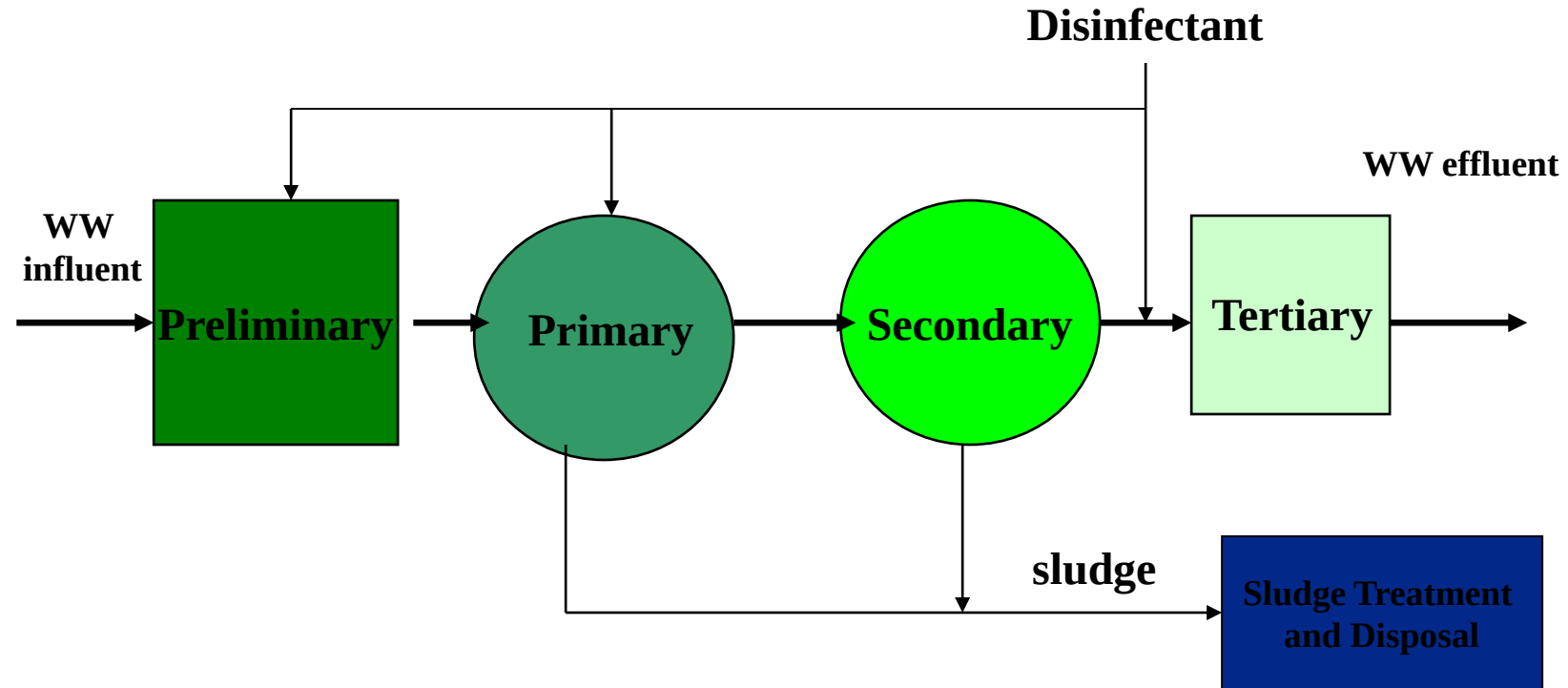


Wastewater treatment

Wastewater Treatment Scheme



Wastewater treatment processes:

- Preliminary treatment is a physical process that removes large contaminants.
- Primary treatment involves physical sedimentation of particulates.
- Secondary treatment involves physical and biological treatment to reduce organic load of wastewater.
- Tertiary or advanced treatments.

Preliminary Treatment

As wastewater enters a treatment facility, it typically flows through a step called preliminary treatment. A screen removes **large floating objects, such as rags, cans, bottles and sticks that may clog pumps, small pipes, and down stream processes.**

- **Coarse screens** remove large solids, rags, and debris from wastewater, and typically have openings of 6 mm (0.25 in) or larger.
- Typical opening sizes for **fine screens** are 1.5 to 6 mm (0.06 to 0.25 in).
- **Very fine screens** with openings of 0.2 to 1.5 mm (0.01 to 0.06 in)

Screens are generally placed in a chamber or channel and inclined towards the flow of the wastewater. The inclined screen allows debris to be caught on the upstream surface of the screen and allows access for manual or mechanical cleaning.

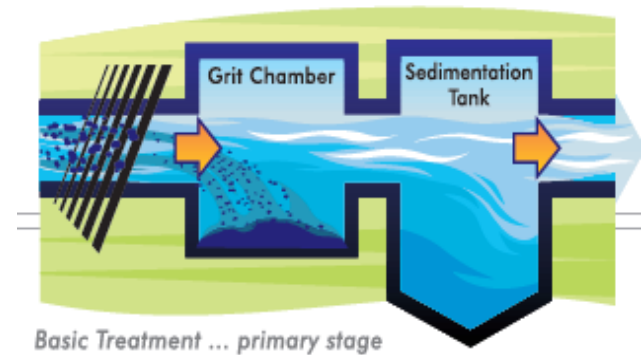
Some plants use devices known as comminutors or barminutors which combine the functions of a screen and a grinder. These devices catch and then cut or shred the heavy solid and floating material. In the process, the pulverized matter remains in the wastewater flow to be removed later in a primary settling tank.

Grit Removal

After the wastewater has been screened, it may flow into a grit chamber where sand, grit, cinders, and small stones settle to the bottom. Removing the grit and gravel that washes off streets or land during storms is very important, especially in cities with combined sewer systems.

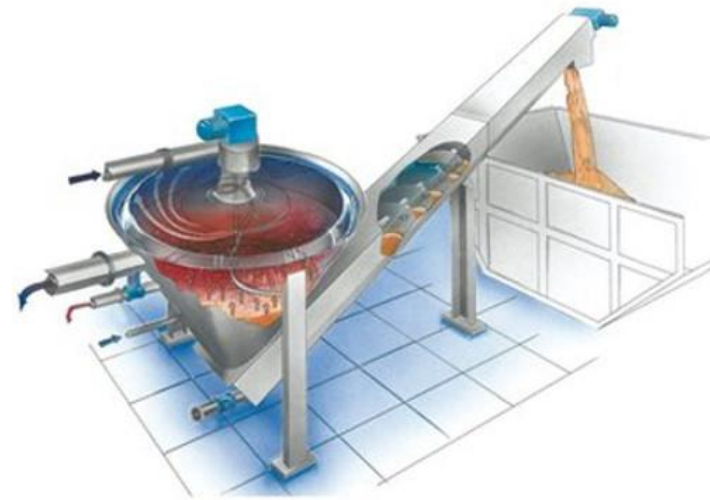
Large amounts of grit and sand entering a treatment plant can cause serious operating problems, such as excessive wear of pumps and other equipment, clogging of aeration devices, or taking up capacity in tanks that is needed for treatment.

In some plants, another finer screen is placed after the grit chamber to remove any additional material that might damage equipment or interfere with later processes. The grit and screenings removed by these processes must be periodically collected and trucked to a landfill for disposal or are incinerated.



Many types of grit removal systems exist, including

- aerated grit chambers,
- vortex-type (paddle or jet induced vortex) grit removal systems,
- detritus tanks (short-term sedimentation basins),
- horizontal flow grit chambers (velocity-controlled channel), and
- hydrocyclones (cyclonic inertial separation).



Primary treatment

- Primary Treatment The initial stage in the treatment of domestic wastewater is known as primary treatment. Coarse solids are removed from the wastewater in the primary stage of treatment. In some treatment plants, primary and secondary stages may be combined into one basic operation. At many wastewater treatment facilities, influent passes through preliminary treatment units before primary and secondary treatment begins.

Primary Sedimentation With the screening completed and the grit removed, wastewater still contains dissolved organic and inorganic constituents along with suspended solids. **The suspended solids consist of minute particles of matter that can be removed from the wastewater with further treatment such as sedimentation or gravity settling, chemical coagulation, or filtration.**

When the wastewater enters a sedimentation tank, it slows down and the suspended solids gradually sink to the bottom. This mass of solids is called **primary sludge**. Various methods have been devised to remove primary sludge from the tanks. Newer plants have some type of mechanical equipment to remove the settled solids from sedimentation tanks. Some plants remove solids continuously while others do so at intervals.



Coagulation and flocculation

They occur in successive steps intended to overcome the forces stabilizing the suspended particles, allowing particle collision and growth of floc.

COAGULATION

The first step destabilizes the particle's charges. Coagulants with charges opposite those of the suspended solids are added to the water to neutralize the negative charges on dispersed non-settlable solids such as clay and color-producing organic substances.

Once the charge is neutralized, the small suspended particles are capable of sticking together. The slightly larger particles, formed through this process and called micro-flocs, are not visible to the naked eye.

A high-energy, rapid-mix to properly disperse the coagulant and promote particle collisions is needed to achieve good coagulation. Over-mixing does not affect coagulation, but insufficient mixing will leave this step incomplete. Coagulants should be added where sufficient mixing will occur. Proper contact time in the rapid-mix chamber is typically 1 to 3 minutes.

FLOCCULATION

Following the first step of coagulation, a second process called flocculation occurs. Flocculation, a gentle mixing stage, increases the particle size from submicroscopic microfloc to visible suspended particles.

The microflocs are brought into contact with each other through the process of slow mixing. Collisions of the microfloc particles cause them to bond to produce larger, visible flocs called pinflocs. The floc size continues to build through additional collisions and interaction with inorganic polymers formed by the coagulant or with organic polymers added. Macroflocs are formed. High molecular weight polymers, called coagulant aids, may be added during this step to help bridge, bind, and strengthen the floc, add weight, and increase settling rate. Once the floc has reached its optimum size and strength, the water is ready for the sedimentation process.

Design contact times for flocculation range from 15 or 20 minutes to an hour or more.

Common Polymers added

Three groups of flocculants are currently used

- MINERAL FLOCCULANTS

- **activated silica.**
- **certain colloidal clays (such as bentonite),**
- **certain metallic hydroxides with a polymeric structure (alum, ferric hydroxide)**

- NATURAL FLOCCULANTS

They are water soluble anionic, cationic or nonionic polymers.

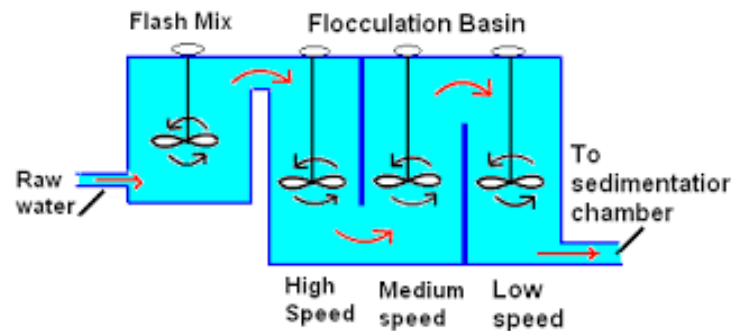
- **the starch derivatives: mostly pregelatinized hence water-soluble. They are corn or potato-starches. They can be natural starches, anionic oxidized starches or amine treated cationic starches.**
- **the polysaccharides: usually guar gums and mostly used in acid medium.**
- **the alginates: anionic and used in potable water treatment.**

- SYNTHETIC FLOCCULANTS

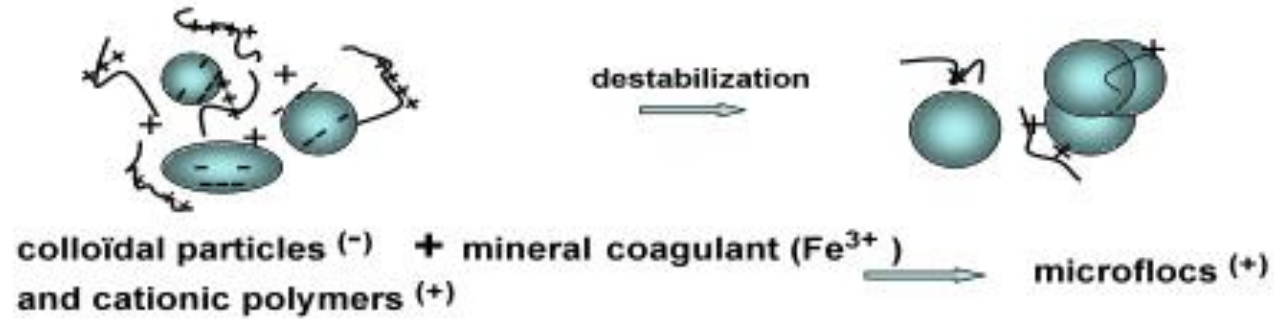
- Polyacrylamides

- The most common polymers are those based on polyacrylamide, which is a nonionic polymer. Their effect is due to bridging between particles by polymer chains.

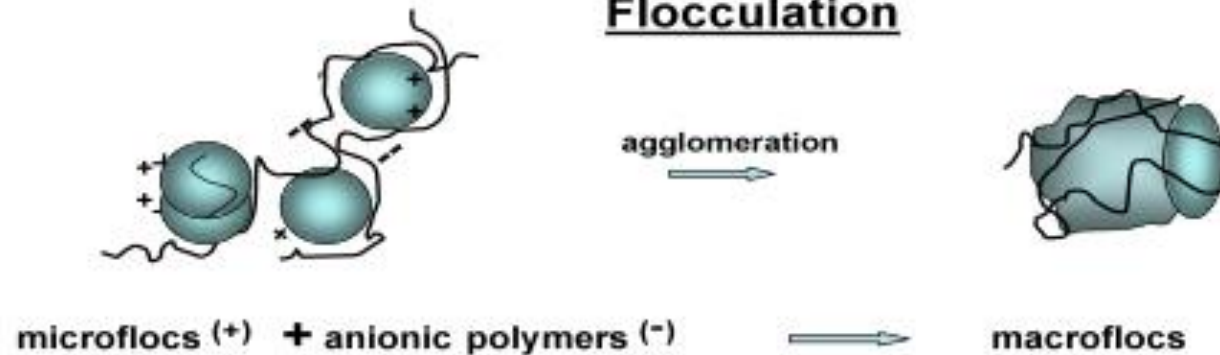
- **Polyethylene-imines**
- **Polyamides-amines**
- **Polyamines**
- **Polyethylene-oxide**
- **Sulfonated compound**



Coagulation



Flocculation



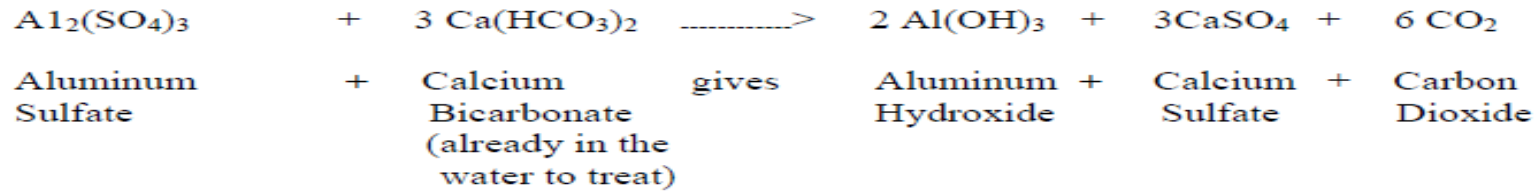
Flocs start to form during the neutralization step as particle collisions occur. ...

Sweep **flocculation** occurs when colloidal contaminants are entrained or swept down by the precipitates as they settle in the suspension. The **optimal pH** range for **coagulation** is 6 to 7 when using alum and 5.5 to 6.5 when using iron.

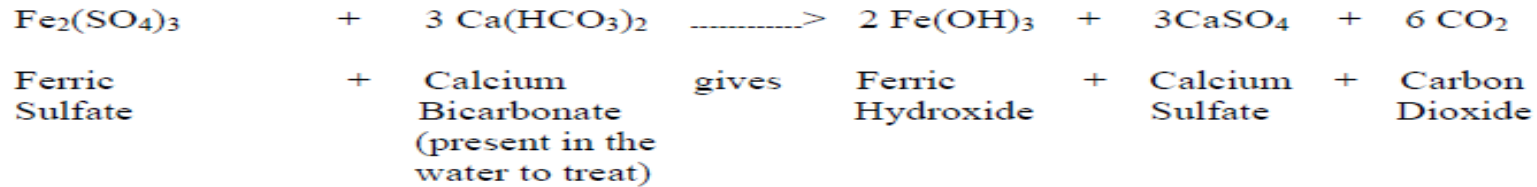
Inorganic Coagulant Reactions

Common coagulant chemicals used are alum, ferric sulfate, ferric chloride, ferrous sulfate, and sodium aluminate. The first four will lower the alkalinity and pH of the solution while the sodium aluminate will add alkalinity and raise the pH. The reactions of each follow:

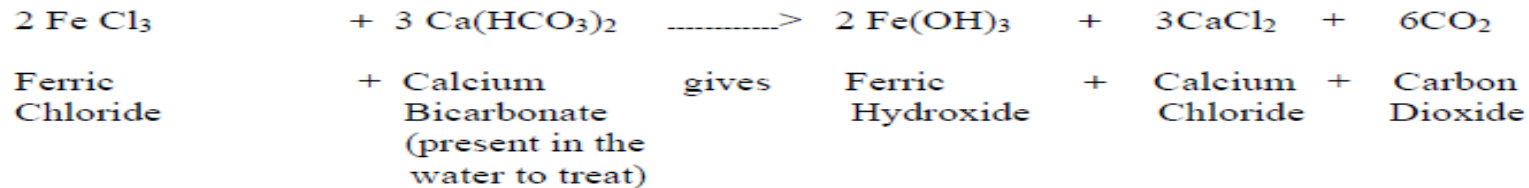
ALUM



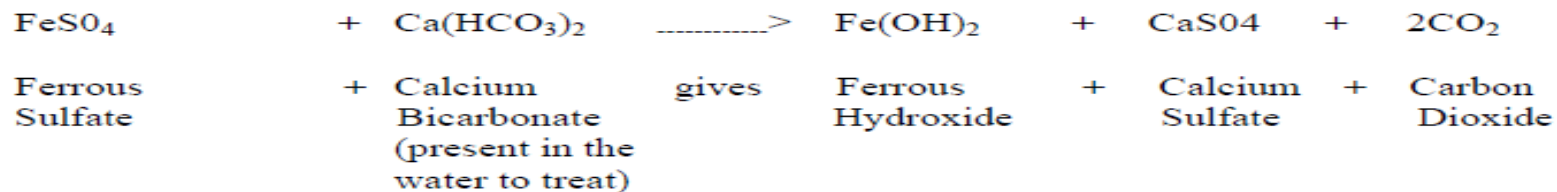
FERRIC SULFATE



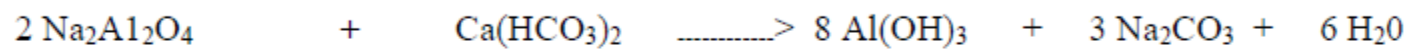
FERRIC CHLORIDE



FERROUS SULFATE



SODIUM ALUMINATE



Sodium Aluminate + Calcium Carbonate (present in the water to treat) gives Aluminum Hydroxide + Sodium Carbonate + Water

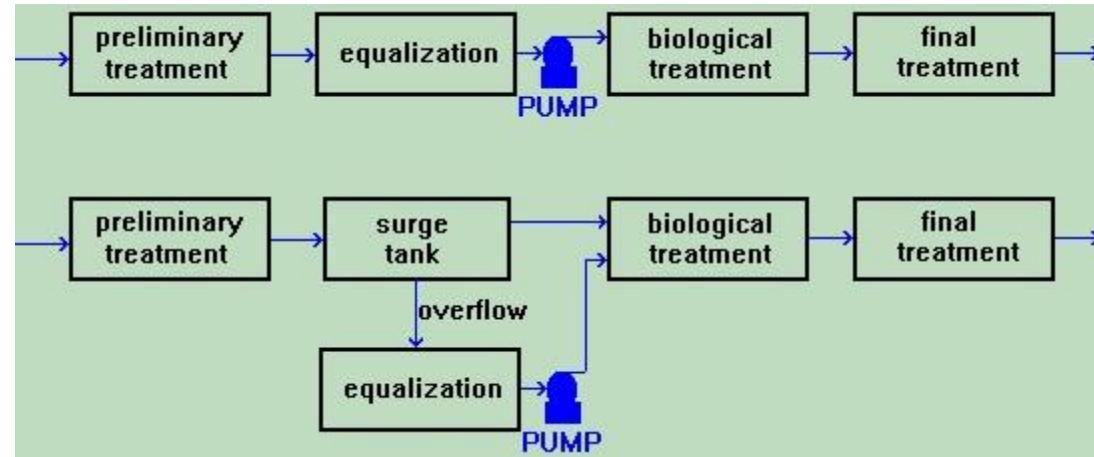


Sodium Aluminate + Carbon Dioxide (present in the water to treat) gives Aluminum Hydroxide + Sodium Carbonate



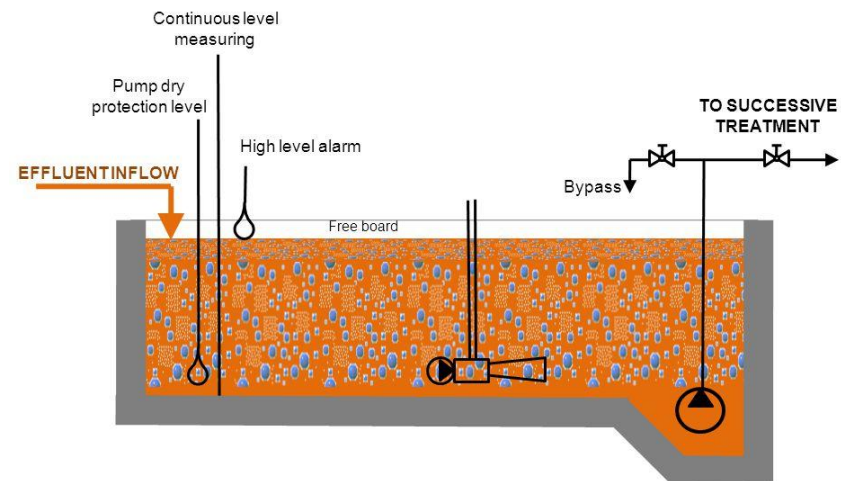
Sodium Aluminate + Magnesium Carbonate (present in the water to treat) gives Magnesium Aluminate + Sodium Carbonate

Equalization



A well - mixed vessel with fluctuating input flow rates and / or concentration with fairly constant output flow rates and/or concentrations.

Equalization – homogenization tank Schematic view



Secondary Treatment

After the wastewater has been through Primary Treatment processes, it flows into the next stage of treatment called secondary. Secondary treatment processes can remove up to 90 percent of the organic matter in wastewater by using biological treatment processes. The two most common conventional methods used to achieve secondary treatment are attached growth processes and suspended growth processes.

If the **BOD/COD ratio** for untreated wastewater is 0.5 or greater, the waste is considered to be easily treatable by biological means. If the **ratio** is below about 0.3, either the waste may have some toxic components or acclimated microorganisms may be required in its stabilization.

Attached Growth Processes

In attached growth (or fixed film) processes, the microbial growth occurs on the surface of stone or plastic media. Wastewater passes over the media along with air to provide oxygen. Attached growth process units include trickling filters, biotowers, and rotating biological contactors. Attached growth processes are effective at removing biodegradable organic material from the wastewater.

A trickling filter is simply a bed of media (typically rocks or plastic) through which the wastewater passes. The media ranges from three to six feet deep and allows large numbers of microorganisms to attach and grow. Older treatment facilities typically used stones, rocks, or slag as the media bed material. New facilities may use beds made of plastic balls, interlocking sheets of corrugated plastic, or other types of synthetic media. This type of bed material often provides more surface area and a better environment for promoting and controlling biological treatment than rock. Bacteria, algae, fungi and other microorganisms grow and multiply, forming a microbial growth or slime layer (biomass) on the media. In the treatment process, the bacteria use oxygen from the air and consume most of the organic matter in the wastewater as food. As the wastewater passes down through the media, oxygen-demanding substances are consumed by the biomass and the water leaving the media is much cleaner. However, portions of the biomass also slough off the media and must settle out in a secondary treatment tank.

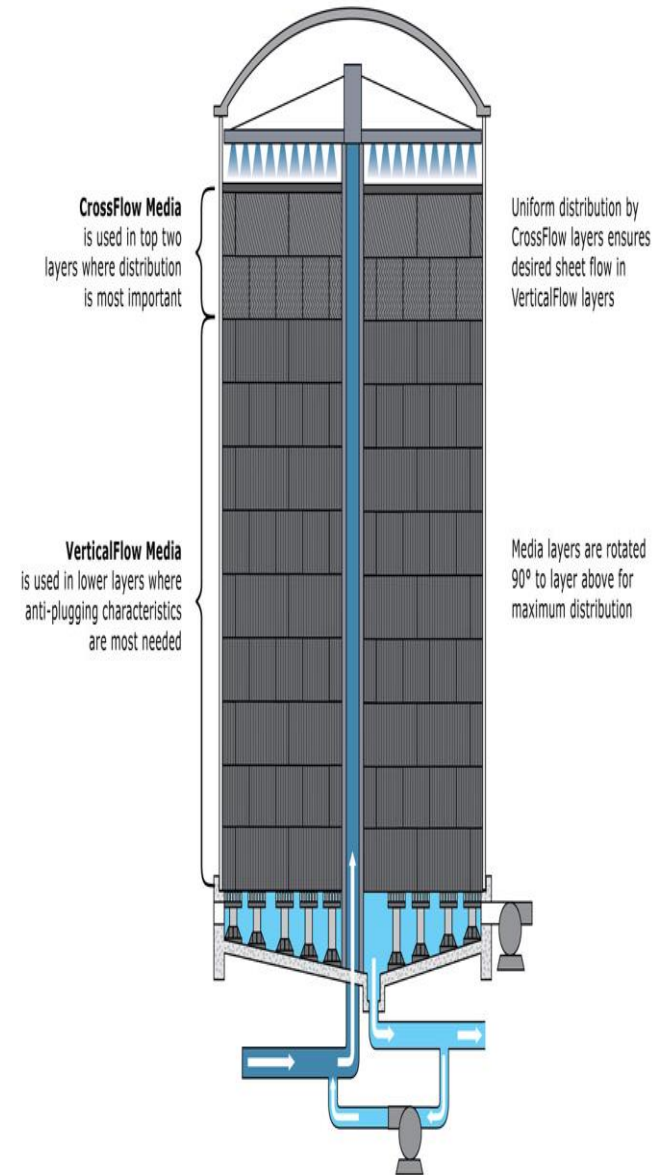


Biotowers

A biotower is used to reduce the BOD value of a liquid effluent. A biotower is an above-ground cylindrical tank or rectangular structure that contains plastic media with a high surface area, such as randomly filled polypropylene shapes or modular blocks of corrugated PVC. Effluent is pumped to the top of the tower and distributed over the surface of the media using rotating distributors, troughs, or nozzles and splash plates. The effluent trickles down over the media, which become coated with microbial films that consume the organic material.

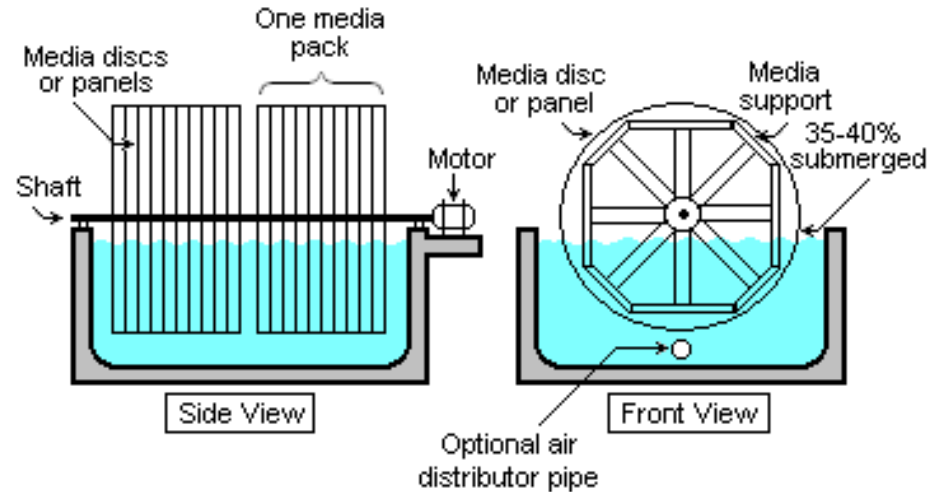
The outer surfaces of the microbial films are highly aerobic, being continuously aerated by the air that passes upwards through the media. Below this the conditions become progressively less aerobic as the film builds up. Eventually the film sloughs off but the plastic medium has a large void volume that ensures unimpeded passage of sloughed biomass, air and liquor.

The treated liquid may be recycled over the biotower before the biological solids are settled out. Biotowers can be arranged in series with inter-stage settlement. Fan ventilation may be incorporated where the biomass must be highly aerobic, for example where nitrification is required.



Rotating biological contactor

- The RBC process involves allowing the wastewater to come in contact with a biological medium in order to remove pollutants in the wastewater before discharge of the treated wastewater to the environment, usually a body of water (river, lake or ocean). A rotating biological contactor is a type of secondary treatment process. It consists of a series of closely spaced, parallel discs mounted on a rotating shaft which is supported just above the surface of the waste water. Microorganisms grow on the surface of the discs where biological degradation of the wastewater pollutants takes place.



The rotating packs of disks (known as the media) are contained in a tank or trough and rotate at between **2 and 5 revolutions per minute**. Commonly used plastics for the media are polyethylene, PVC and expanded polystyrene. The shaft is aligned with the flow of wastewater so that the discs rotate at **right angles to the flow** with several packs usually combined to make up a treatment train. About **40% of the disc area is immersed in the wastewater**.

Biological growth is attached to the surface of the disc and forms a slime layer. The discs contact the wastewater with the atmospheric air for oxidation as it rotates. The rotation helps to slough off excess solids. The disc system can be staged in series to obtain nearly any detention time or degree of removal required. Since the systems are staged, the culture of the later stages can be acclimated to the slowly degraded materials.

The discs consist of plastic sheets ranging **from 2 to 4 m in diameter and are up to 10 mm thick**. Several modules may be arranged in parallel and/or in series to meet the flow and treatment requirements. The discs are submerged in waste water to about 40% of their diameter. Approximately 95% of the surface area is thus alternately submerged in waste water and then exposed to the atmosphere above the liquid. Carbonaceous substrate is removed in the initial stage of RBC. Carbon conversion may be completed in the first stage of a series of modules, with nitrification being completed after the 5th stage. Most design of RBC systems will include a minimum of 4 or 5 modules in series to obtain nitrification of waste water.

Biofilms, which are biological growths that become attached to the discs, assimilate the organic materials in the wastewater. Aeration is provided by the rotating action, which exposes the media to the air after contacting them with the wastewater, facilitating the degradation of the pollutants being removed. The degree of wastewater treatment is related to the amount of media surface area and the quality and volume of the inflowing wastewater.

Suspended growth process

Suspended Growth Processes Similar to the microbial processes in attached growth systems, suspended growth processes are designed to remove biodegradable organic material and organic nitrogen-containing material by converting ammonia nitrogen to nitrate unless additional treatment is provided. In suspended growth processes, the microbial growth is suspended in an aerated water mixture where the air is pumped in, or the water is agitated sufficiently to allow oxygen transfer.

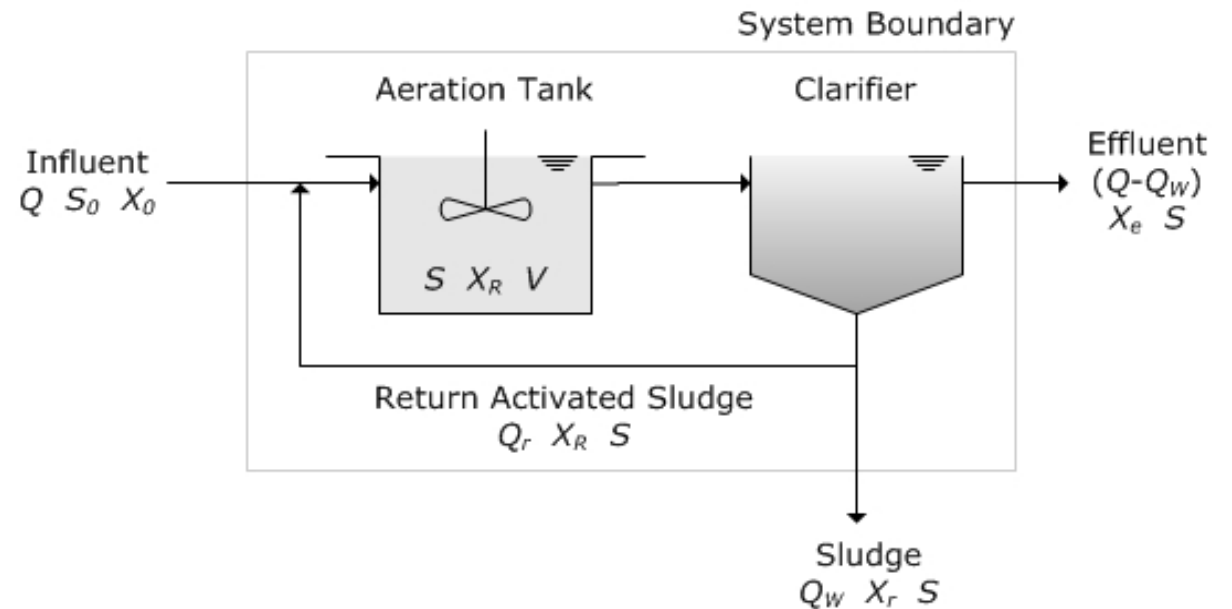
The suspended growth process speeds up the work of aerobic bacteria and other microorganisms that break down the organic matter in the sewage by providing a rich aerobic environment where the microorganisms suspended in the wastewater can work more efficiently. In the aeration tank, wastewater is vigorously mixed with air and microorganisms acclimated to the wastewater in a suspension for several hours. This allows the bacteria sequencing and other microorganisms to break down the organic matter in the wastewater. The microorganisms grow in number and the excess biomass is removed by settling before the effluent is discharged or treated further. Now activated with millions of additional aerobic bacteria, some of the biomass can be used again by returning it to an aeration tank for mixing with incoming wastewater.

ACTIVATED SLUDGE TREATMENT

- ❑ It basically consists of an *aeration tank* and a *secondary settling basin or clarifier*.
- ❑ In aeration tank, microorganisms consume the dissolved organic pollutants as food.
- ❑ The aerobic microorganisms in the tank grow and multiply forming a suspension of biological solids called activated sludge.
- ❑ The combination of activated sludge and wastewater in aeration tank is called mixed liquor.
- ❑ A tank detention time of about 6hrs is required for stabilization of most of organics in mixed liquor.
- ❑ The mixed liquor flows to the secondary or final clarifier ,in which activated sludge solids settles out by gravity.

The activated sludge process, like most other techniques, has advantages and limitations. The units necessary for this treatment are relatively small, requiring less space than attached growth processes. In addition, when properly operated and maintained, the process is generally free of flies and odors. However, most activated sludge processes are more costly to operate than attached growth processes due to higher energy use to run the aeration system. The effectiveness of the activated sludge process can be impacted by elevated levels of toxic compounds in wastewater unless complex industrial chemicals are effectively controlled through an industrial pretreatment program.

Typical retention time = 3-8 hrs depending on BD load



An adequate supply of oxygen is necessary for the activated sludge process to be effective. The oxygen is generally supplied by mixing air with the sewage and biologically active solids in the aeration tanks by one or more of several different methods. Mechanical aeration can be accomplished by drawing the sewage up from the bottom of the tank and spraying it over the surface, thus allowing the sewage to absorb large amounts of oxygen from the atmosphere. Pressurized air can be forced out through small openings in pipes suspended in the wastewater. Combination of mechanical aeration and forced aeration can also be used. Also, relatively pure oxygen, produced by several different manufacturing processes, can be added to provide oxygen to the aeration tanks.

From the aeration tank, the treated wastewater flows to a sedimentation tank (secondary clarifier), where the excess biomass is removed. Some of the biomass is recycled to the head end of the aeration tank, while the remainder is “wasted” from the system. The waste biomass and settled solids are treated before disposal or reuse as biosolids.

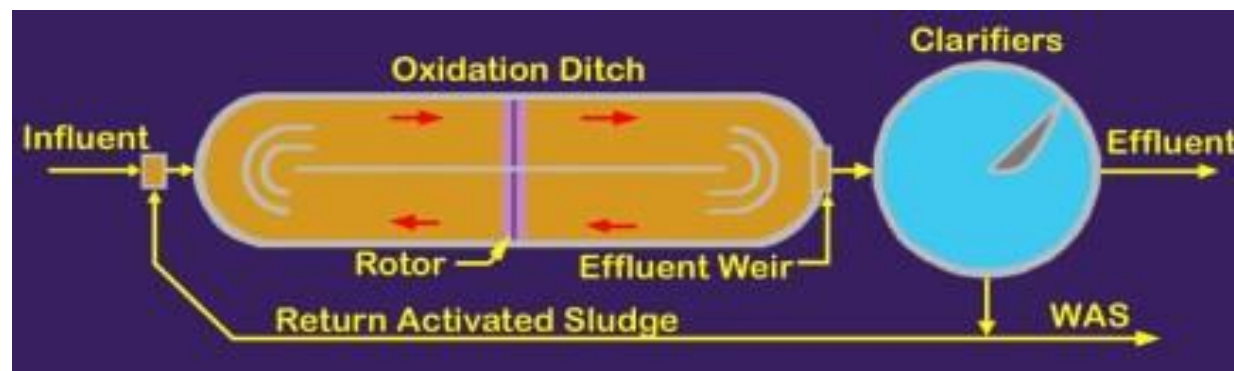
Description of Oxidation Ditch

Oxidation ditch is an extended aeration activated sludge process. An oxidation ditch is a large holding tank in a continuous ditch with oval shape similar to that of a racetrack. The ditch is built on the surface of the ground and is lined with an impermeable lining. This allows the waste water to have plenty of exposure to the open air for the diffusion of oxygen.

The liquid depth in the ditches is very shallow, 0.9 to 1.5 m, which helps to prevent anaerobic conditions from occurring at the bottom of the ditch.

The oxidation ditch effluent is clarified in a secondary clarifier and the settled sludge is returned to maintain a desirable MLSS concentration. The MLSS concentration in the oxidation ditch generally ranges from 3,000 mg/ L to 5,000 mg/ L; however, this is dependent upon the surface area provided for sedimentation, the rate of sludge return, and the aeration process.

Longer retention time within the ditch will allow for a greater amount of organic matter to be broken down by the aerobic bacteria. After treatment, the waste water is pumped to a secondary settling tank where the sludge and the water are allowed to separate. From there the effluent goes on to other treatment processes or disposal. The sludge that has accumulated on the bottom of the secondary settling tank is then removed and a portion of it is returned to the ditch to facilitate microbial activity in the next batch of sewage to be treated.



MLSS is used to indicate the concentration of suspended solids in activated sludge.

MLVSS represents the concentration of biomass in activated sludge.

To measure the MLVSS, you can filtrate the activated sludge through 0.45 micro filter. Then, heating at 550 degree C



$$\text{VSS, mg/L} = \frac{\text{Weight of Material Lost by Burning (g)}}{\text{sample volume, mL}}$$

Food to Microorganism Ratio F/M Activated Sludge Systems

The F/M ratio is a process control number that helps you to determine the proper number of microorganisms for your system. To do this calculation, you will need the following information:

- Influent Flow into your activated sludge system (Flow MGD)
- Influent CBOD (mg/l) concentration into your aeration tank.
- Mixed Liquor Volatile Suspended Solids Concentration (mg/l)
- Volume (in gallons) of your aeration system

To determine the amount of incoming food (**F**), you need to know the CBOD of the influent into your activated sludge(aeration) system. You also need to know the flow(MGD). So to calculate the amount of food we do the following calculation:

$$\mathbf{F = Influent Flow (MGD) \times Influent CBOD Concentration (mg/l) \times 8.34}$$

To determine the volume of microorganisms (**M**), you need to know the volume of your aeration system and you need to know the concentration of Volatile Solids in your aeration system (MLVSS) or Mixed Liquor Volatile Suspended Solids.

To calculate the microorganisms, we do the following calculation:

$$\mathbf{M = Aeration\ System\ Volume\ (in\ Millions\ of\ Gallons)\ X\ MLVSS\ X\ 8.34}$$

To Calculate the Ratio: F/M

Example:

Facility Flow = 1.2 MGD Influent CBOD= 230 mg/l

$$1.2 \times 230 \times 8.34 = 2301.84 \text{ Lbs Coming In}$$

$$\mathbf{F = 2301}$$

Aeration System Volume 250,000 gal / 1,000,000 = 0.25MG

MLVSS = 2500mg/l

$$0.25 \times 2500 \times 8.34 = 5215 \text{ Lbs of Microorganisms under Aeration}$$

$$\mathbf{M = 5212}$$

$$\mathbf{\underline{2301/5212 = 0.44\ F/M\ Ratio}}$$

The optimal range for the **F:M ratio** in a complete mix **system** is 0.2 to 0.6 lb BOD/lb MLVSS (kg BOD/kg MLVSS)